

Data Science Interview Questions

Questions Only

1. Data Science Fundamentals

1. What is Data Science?
2. What is the difference between Data Science, Data Analytics, and Machine Learning?
3. What is the Data Science lifecycle?
4. What are the main steps in a Data Science project?
5. What is exploratory data analysis (EDA)?
6. Why is EDA important?
7. What is descriptive analysis?
8. What is diagnostic analysis?
9. What is predictive analysis?
10. What is prescriptive analysis?

2. Python for Data Science

11. Why is Python widely used in Data Science?
12. What are NumPy and Pandas?
13. What is a NumPy array?
14. What is the difference between a Python list and a NumPy array?
15. What is a Pandas Series?
16. What is a Pandas DataFrame?
17. How do you read a CSV file using Pandas?
18. How do you inspect the first and last rows of a DataFrame?
19. How do you check missing values in Pandas?
20. How do you remove duplicate rows?
21. How do you filter rows in a DataFrame?
22. What is groupby() in Pandas?
23. What is merge() in Pandas?
24. What is concat() in Pandas?
25. What is apply() in Pandas?

3. NumPy and Pandas

- 26. What is vectorization in NumPy?
- 27. What is broadcasting in NumPy?
- 28. What is reshaping an array?
- 29. What is the difference between loc and iloc?
- 30. What is the difference between merge, join, and concat?
- 31. How do you sort a DataFrame?
- 32. How do you rename columns?
- 33. How do you change a column's data type?
- 34. How do you create a new column from existing columns?
- 35. How do you aggregate data in Pandas?
- 36. How do you handle categorical columns?
- 37. How do you detect outliers using Pandas?

4. Statistics

- 38. What is statistics?
- 39. What is the difference between population and sample?
- 40. What is mean?
- 41. What is median?
- 42. What is mode?
- 43. What is variance?
- 44. What is standard deviation?
- 45. What is covariance?
- 46. What is correlation?
- 47. What is the difference between correlation and covariance?
- 48. What is probability?
- 49. What is conditional probability?
- 50. What is Bayes' theorem?
- 51. What is a normal distribution?
- 52. What is the central limit theorem?
- 53. What is a confidence interval?
- 54. What is hypothesis testing?
- 55. What is a p-value?
- 56. What is a null hypothesis?

- 57. What is a Type I error?
- 58. What is a Type II error?

5. Data Visualization

- 59. Why is data visualization important?
- 60. What is Matplotlib?
- 61. What is Seaborn?
- 62. When would you use a bar chart?
- 63. When would you use a histogram?
- 64. When would you use a box plot?
- 65. When would you use a scatter plot?
- 66. When would you use a line chart?
- 67. What does a heatmap show?
- 68. How do you visualize correlations?
- 69. How can you avoid misleading visualizations?

6. Machine Learning for Data Scientists

- 70. What is supervised learning?
- 71. What is unsupervised learning?
- 72. What is regression?
- 73. What is classification?
- 74. What is clustering?
- 75. What is feature engineering?
- 76. What is feature selection?
- 77. What is dimensionality reduction?
- 78. What is overfitting?
- 79. What is underfitting?
- 80. What is cross-validation?
- 81. What is model evaluation?

7. SQL and Databases

- 82. What is SQL?
- 83. What is a database?

- 84. What is a relational database?
- 85. What is a primary key?
- 86. What is a foreign key?
- 87. What is normalization?
- 88. What is denormalization?
- 89. What is an INNER JOIN?
- 90. What is a LEFT JOIN?
- 91. What is GROUP BY?
- 92. What is HAVING?
- 93. What is a subquery?
- 94. What is a CTE?
- 95. What is a window function?

8. Data Cleaning and Business Problems

- 96. How would you approach a new dataset?
- 97. How would you identify data quality issues?
- 98. How would you handle inconsistent categorical values?
- 99. How would you handle missing data in a business dataset?
- 100. How would you identify duplicate customers?
- 101. How would you investigate an unexpected drop in sales?
- 102. How would you identify the most important factors affecting customer churn?
- 103. How would you communicate analytical findings to a non-technical stakeholder?

9. Project-Based Data Science Questions

- 104. Explain your Data Science project.
- 105. What business problem did your project solve?
- 106. How did you collect the data?
- 107. How did you clean the data?
- 108. What EDA did you perform?
- 109. What features did you create?
- 110. Which model did you choose and why?
- 111. How did you evaluate the model?
- 112. What challenges did you face?

113. How did you deploy or present your results?

114. What would you improve in your project?